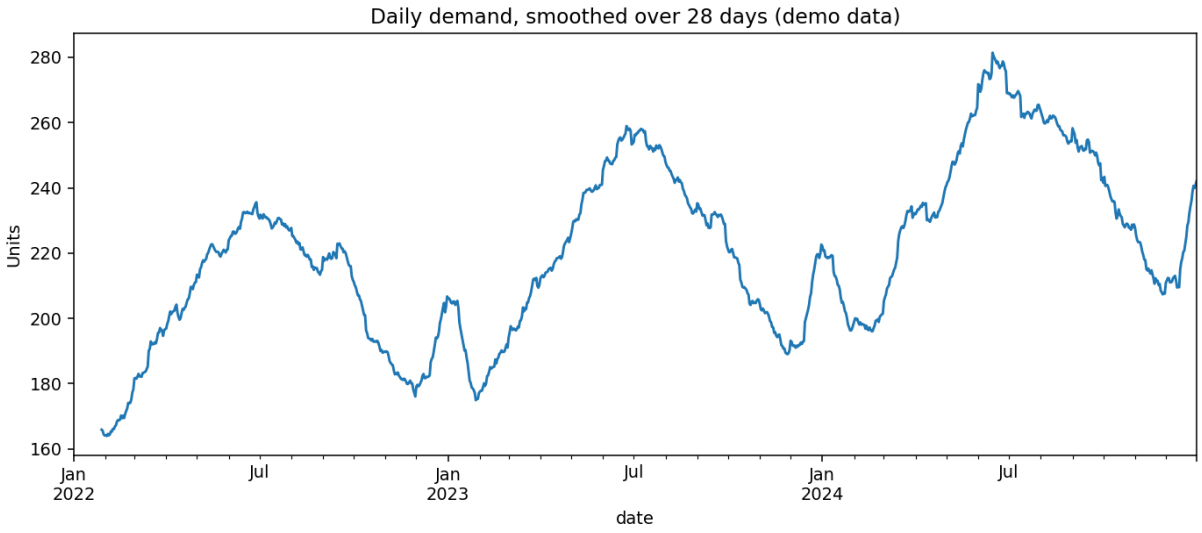
Retail Demand Forecasting & Inventory Review

A planning case study built with synthetic demo data

Executive summary

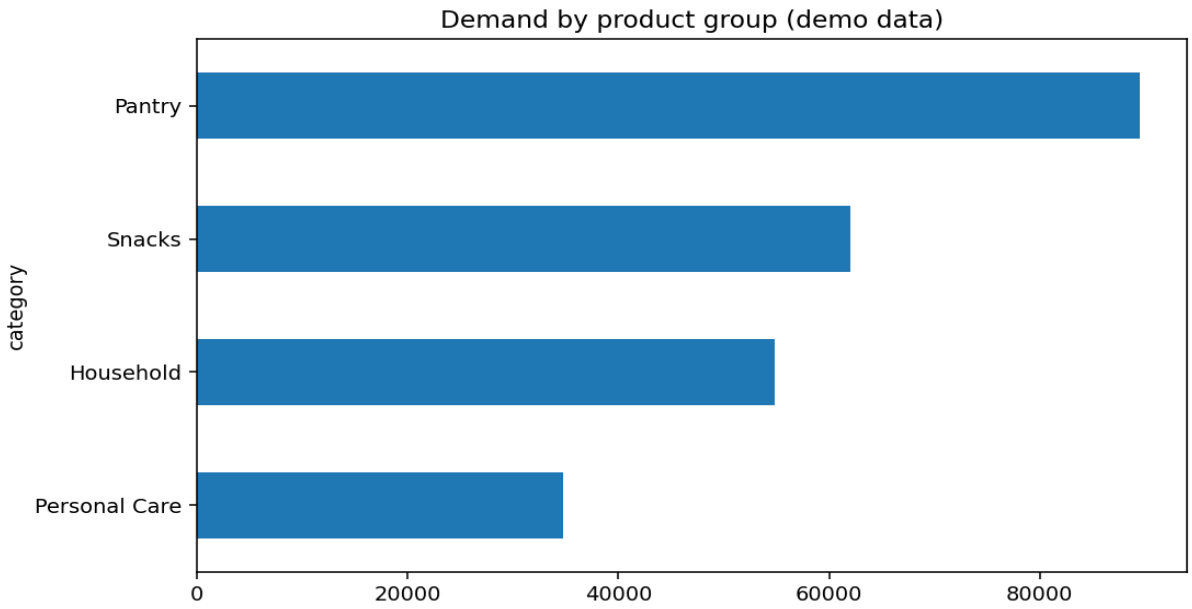
This project connects demand patterns to inventory decisions. It compares simple forecast baselines, estimates near-term demand and builds a reorder list using demand, lead time and safety stock. The numbers in this report are generated for demonstration and are not retailer results.

Demo file: 17,536 daily observations; 241,176 observed units; 8 SKUs with a positive suggested reorder.



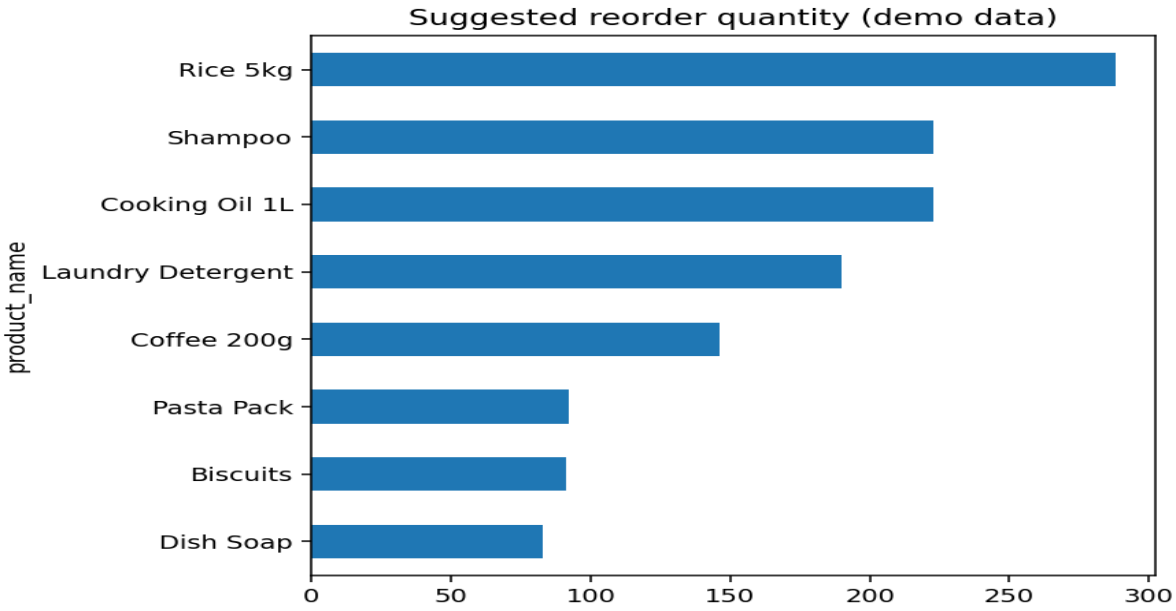
Forecasting approach

The demo compares trailing 7-day and 28-day average baselines against a later 60-day holdout. This is a basic starting point, not the end of the model selection process. A production analysis should add seasonal-naive forecasts, rolling-origin backtests, weighted absolute percentage error and forecast bias by SKU and store.



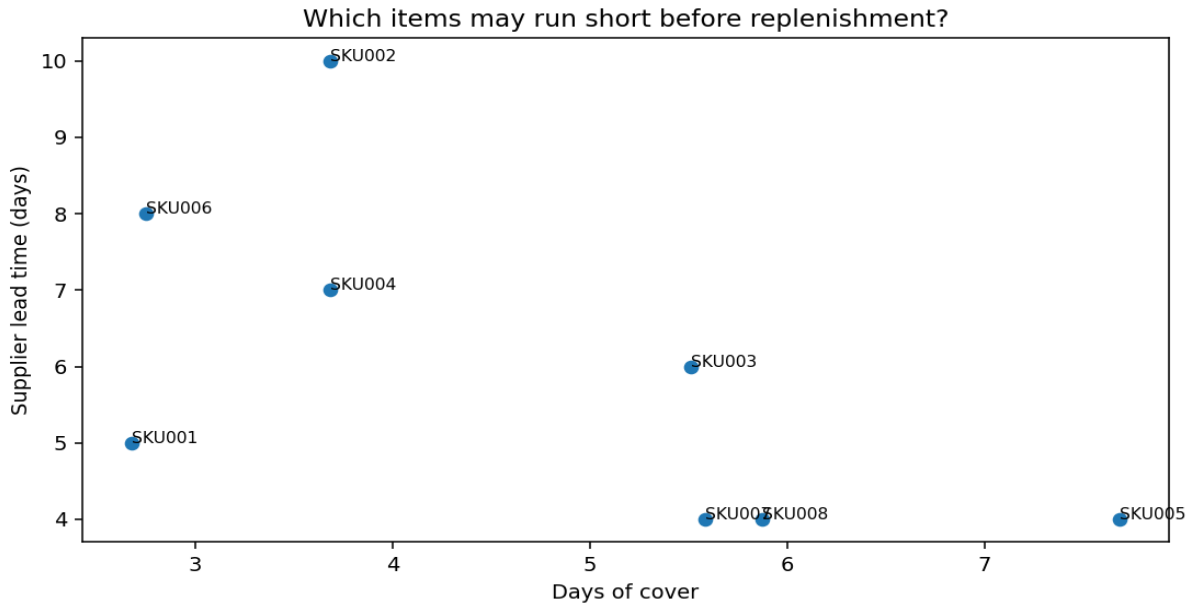
Inventory policy

The reorder worksheet estimates target stock from average daily demand across lead time and review period, adds a safety-stock allowance for recent variation, and subtracts current on-hand stock. It does not account for open purchase orders, case packs, expiry, supplier minimums or every source of stockout-censored demand.



The less obvious problem: sales are not always demand

When an item is unavailable, recorded sales can fall even if customers still want it. The demo includes a stockout proxy to make this issue visible. Real analysis should use availability logs and inventory snapshots to identify censored demand rather than treating every zero as weak demand.



Recommended next steps

1. Begin with high-volume items and compare a seasonal-naive baseline with moving averages.
2. Review items whose cover is below supplier lead time before placing orders.
3. Add inbound purchase orders, pack sizes, supplier minimums and shelf-life constraints.
4. Monitor forecast error and bias each week, with promotions and stockouts reviewed separately.
5. Test how reorder recommendations change when lead times or service targets change.

Data note

The included CSV files are synthetic. The Walmart M5 Forecasting Accuracy dataset is available through Kaggle at https://www.kaggle.com/competitions/m5-forecasting-accuracy/data. It is not included in this archive; download it under the current competition terms and follow DATA_GUIDE.md to prepare it.